Day 6 — TNPSC Group 2A Study Material

Part A – General Studies: Elements, Compounds, Acids, Bases, Salts, Petroleum Products, Fertilizers & Pesticides

READING MATERIAL

This clause covers a lot of distinct chemistry content (elements/compounds, acids/bases/salts, petroleum, fertilizers, pesticides). Real exam evidence shows TNPSC favors comparison/matching formats here (conjugate acid pairs, fertilizer-type matching, ore-mineral pairing) over plain definition recall.

Q: Real exam: In each conjugate acid-base pair, which member is the stronger acid — the protonated form or the deprotonated form? Apply to H_3O^+/H_2O, NH_4^+/NH_3, H_2S/HS^-, H_2O/OH^-.

A: The protonated (acid) form is always the stronger acid in each pair: H_3O^+ (stronger than H_2O), NH_4^+ (stronger than NH_3), H_2S (stronger than HS^-), H_2O (stronger than OH^-). General rule: within a conjugate pair, the species that CAN donate a proton is the acid, and it's stronger than the species left behind after donating it.

Q: Real exam (Assertion-Reason): "Acid rain means the presence of acids in rain water" — why does this happen?

A: Because the atmosphere contains oxides of nitrogen and sulphur produced by burning fuels (combustion). These oxides dissolve in atmospheric water vapour to form nitric and sulphuric acid, which fall as 'acid rain'. (Marked in a real exam as: both statements true, and the reason correctly explains the assertion.)

Q: Real exam: Which organic pesticide contains phosphorus?

A: Parathion. (DDT, BHC, and 2,4-D — the other options in that question — do not contain phosphorus.)

Q: Real exam: Match each fertilizer type to its example — Mixed fertilizer, Complex fertilizer, Bio fertilizer, Organic nitrogen fertilizer.

A: Mixed fertilizer → NPK (a blend of Nitrogen, Phosphorus, Potassium). Complex fertilizer → DAP (Di-Ammonium Phosphate, a single chemical compound providing multiple nutrients). Bio fertilizer → Algae (living organisms that enrich soil). Organic nitrogen fertilizer → Oil cake (a plant-derived byproduct).

Q: Real exam: Which ore-mineral pairings are correct, and which is the deliberately wrong one? Hematite Ore–Oxide of iron; Magnetite Ore–Black ore; Limonite–Dolomite; Siderite–Iron Carbonate.

A: Correct: Hematite–Oxide of iron, Magnetite–Black ore (magnetite literally is a black-coloured iron oxide ore), Siderite–Iron Carbonate. WRONG: Limonite–Dolomite — limonite is actually a hydrated iron oxide, not dolomite (dolomite is a calcium-magnesium carbonate mineral, unrelated to limonite).

Q: What are the defining properties of acids (taste, litmus behaviour, ion produced)?

A: Sour taste, turns blue litmus paper red, produces H^+ (or H_3O^+) ions in water, and conducts electricity in solution.

Q: What are the defining properties of bases (taste, litmus behaviour, ion produced)?

A: Bitter taste, turns red litmus paper blue, produces OH^- ions in water, and conducts electricity in solution. A base that is soluble in water is specifically called an alkali.

Q: What is the pH scale's range, and what do the values 0, 7, and 14 represent?

A: Range 0-14. 0 = strongly acidic, 7 = neutral (pure water), 14 = strongly basic/alkaline. Acid rain typically has a pH below 5.6.

Q: Match common food acids to their source: Acetic acid, Citric acid, Lactic acid, Tartaric acid.

A: Acetic acid → vinegar. Citric acid → citrus fruits (lemon, orange). Lactic acid → curd. Tartaric acid → tamarind.

Q: What are common everyday uses of these salts: bleaching powder, sodium hydrogen carbonate, sodium carbonate?

A: Bleaching powder → disinfectant/oxidizing agent (and water treatment, along with chlorine). Sodium hydrogen carbonate (baking soda) → antacid and cooking. Sodium carbonate (washing soda) → cleaning.

MOCK TEST (WITH ANSWERS)

1. In the conjugate pair H_2S / HS^-, which is the stronger acid?

A. H_2S ✓

B. HS^-

C. Both are equal

D. Neither is an acid

Explanation: The protonated form (H_2S) is the stronger acid of the conjugate pair. [medium, question-bank-2022]

2. Acid rain forms primarily because the atmosphere contains oxides of which two elements, produced by burning fuels?

A. Carbon and Hydrogen

B. Nitrogen and Sulphur ✓

C. Oxygen and Helium

D. Calcium and Magnesium

Explanation: Oxides of nitrogen and sulphur dissolve in atmospheric moisture to form acids. [medium, question-bank-2025]

3. Which of these organic pesticides contains phosphorus?

A. DDT

B. BHC

C. 2,4-D

D. Parathion ✓

Explanation: Parathion is a phosphorus-containing (organophosphate) pesticide. [medium, question-bank-2017]

4. Which fertilizer type is best represented by NPK?

A. Bio fertilizer

B. Complex fertilizer

C. Mixed fertilizer ✓

D. Organic nitrogen fertilizer

Explanation: Mixed fertilizer is exemplified by NPK blends. [medium, question-bank-2017]

5. Which of the following ore-mineral pairings is INCORRECT?

A. Hematite – Oxide of iron

B. Magnetite – Black ore

C. Limonite – Dolomite ✓

D. Siderite – Iron Carbonate

Explanation: Limonite is a hydrated iron oxide, not dolomite — dolomite is an unrelated calcium-magnesium carbonate mineral. [hard, question-bank-2024]

6. Acids turn blue litmus paper which colour?

A. Green

B. Red ✓

C. Yellow

D. Stays blue

Explanation: Acids turn blue litmus red. [easy, web-research]

7. Which ion is characteristically produced by bases in water?

A. H^+

B. OH^- ✓

C. Cl^-

D. Na^+

Explanation: Bases produce OH^- ions in aqueous solution. [easy, web-research]

8. On the pH scale (0-14), a value of 7 represents:

A. Strongly acidic

B. Strongly basic

C. Neutral ✓

D. Undefined

Explanation: pH 7 = neutral (pure water). [easy, web-research]

9. Which acid is found in tamarind?

A. Citric acid

B. Lactic acid

C. Tartaric acid ✓

D. Acetic acid

Explanation: Tartaric acid. [easy, web-research]

10. Washing soda (sodium carbonate) is primarily used for:

A. Cooking

B. Cleaning ✓

C. Disinfecting wounds

D. Fertilizing crops

Explanation: Cleaning — washing soda is a common household cleaning agent. [easy, web-research]

Part B – Aptitude & Mental Ability: Simple Interest

READING MATERIAL

Simple Interest is 1 of 10 named Aptitude sub-skills. Real exam evidence shows TNPSC favors word-problem variants (doubling time, comparing two loans, fractional rates with month-based time) over the bare 'calculate SI' formula alone.

Q: Formula: What is the Simple Interest formula, and how do you get the doubling time from it?

A: SI = (Principal × Rate × Time) / 100. A sum DOUBLES when SI equals the Principal itself: P = (P×R×T)/100 → T = 100/R years. So at 8% per annum, doubling time = 100/8 = 12.5 years = 150 months.

Q: Real exam (2024): In how many months will Rs.2,000 double itself at 8% per annum simple interest?

A: T = 100/R = 100/8 = 12.5 years = 12.5 × 12 = 150 months.

Q: Real exam (2022): What is the simple interest on Rs.1,000 for one year at 10% per annum?

A: SI = (1000 × 10 × 1)/100 = Rs.100.

Q: Real exam (2024): Find the simple interest on Rs.68,000 at 16⅔% per annum for 9 months.

A: Convert the mixed-fraction rate: 16⅔% = 50/3 %. Convert months to years: 9 months = 9/12 = 3/4 year. SI = 68,000 × (50/3) × (3/4) / 100 = 68,000 × 12.5/100 = Rs.8,500. (The 3s cancel neatly between the rate's denominator and the time's numerator — recognizing that shortcut avoids messy fraction arithmetic.)

Q: Real exam (2025): A borrows Rs.16,000 at 12% p.a. simple interest, B borrows Rs.12,000 at 18% p.a. simple interest. In how many years will their total amounts owed (principal+interest) become equal?

A: Set up the equality: 16000 + 16000(0.12)t = 12000 + 12000(0.18)t → 16000+1920t = 12000+2160t → 4000 = 240t → t = 16⅔ years.

Q: Method — finding the rate when given principal, time, and final amount: A sum of Rs.5,000 becomes Rs.6,200 in 3 years at simple interest. Find the rate.

A: SI = Amount – Principal = 6200–5000 = 1200. Rearranged formula: R = (SI×100)/(P×T) = (1200×100)/(5000×3) = 120000/15000 = 8% per annum.

MOCK TEST (WITH ANSWERS)

1. In how many years will a sum double itself at 5% per annum simple interest?

A. 10 years

B. 15 years

C. 20 years ✓

D. 25 years

Explanation: T = 100/R = 100/5 = 20 years. [easy, authored]

2. Find the simple interest on Rs.2,500 for 1 year at 12% per annum.

A. Rs.250

B. Rs.275

C. Rs.300 ✓

D. Rs.320

Explanation: SI = (2500×12×1)/100 = Rs.300. [easy, authored]

3. Find the simple interest on Rs.54,000 at 13⅓% per annum for 6 months.

A. Rs.3,600 ✓

B. Rs.3,000

C. Rs.3,300

D. Rs.4,000

Explanation: 13⅓% = 40/3%. 6 months = 1/2 year. SI = 54000×(40/3)×(1/2)/100 = 54000×40/600 = 54000×(1/15) = 3,600. [hard, authored]

4. C borrows Rs.20,000 at 10% p.a. simple interest, D borrows Rs.15,000 at 15% p.a. simple interest. In how many years will their total amounts owed become equal?

A. 8 years

B. 10 years

C. 20 years ✓

D. 25 years

Explanation: 20000+2000t = 15000+2250t → 5000 = 250t → t = 20 years. [hard, authored]

5. A sum of Rs.8,000 becomes Rs.9,440 in 3 years at simple interest. Find the rate per annum.

A. 5%

B. 6% ✓

C. 6.5%

D. 7%

Explanation: SI = 9440−8000 = 1440. R = (1440×100)/(8000×3) = 144000/24000 = 6%. [medium, authored]

6. What sum will yield Rs.900 as simple interest in 3 years at 6% per annum?

A. Rs.4,000

B. Rs.5,000 ✓

C. Rs.5,500

D. Rs.6,000

Explanation: P = (SI×100)/(R×T) = (900×100)/(6×3) = 90000/18 = Rs.5,000. [medium, authored]

Part C – General English: Poem: "The Age of Chivalry" — George Krokos

READING MATERIAL

A poem (from the Samacheer Kalvi Class 7/10 English anthology, depending on edition) celebrating knightly adventure, romance, and idealism. Real coaching-site evidence shows TNPSC tests figures of speech, the poem's theme, AND in-context vocabulary meaning roughly equally — cover all three.

Q: What is the poem about, in brief?

A: Knights in the Age of Chivalry were strong, bold, and carried swords/lances, covering themselves in shining armour and seeking romance. They fought for righteous causes, displayed valour, and won their lady-love's affection through adventures aimed at goodness and sacrifice. The poem's main theme is Adventure and Chivalry.

Q: What did knights mainly fight for, according to the poem?

A: Show, love, and glory — not money, revenge, or peace.

Q: What is the poet's view on imagination, per this poem?

A: Imagination leads to adventure — it's portrayed positively, as the spark for adventurous, chivalric ideals, not as something that 'causes problems' or 'blocks reality'.

Q: What does "shining armour" represent in the poem?

A: Glory and honour — not literally just metal or wealth.

Q: Identify the figure of speech: "adventures hide themselves" and "abated breath."

A: Both are cited as Personification — giving 'adventures' a human-like ability to 'hide', and 'breath' the ability to be 'abated' (held back), as if it were acting with intent.

Q: Identify the figure of speech: "Maybe life itself is one big adventure."

A: Metaphor — life is directly equated with 'one big adventure', without using 'like/as'.

Q: In context, what does "you bounce a bit" convey, and what does "transparent" mean here?

A: "You bounce a bit" conveys excitement (a physical sign of eager energy). "Transparent" in this poem means being open to experience, not literally see-through.

Q: In context, what do "embrace you" and "days of yore" mean?

A: "Embrace you" means to welcome closely. "Days of yore" refers to ancient times (the historical setting of knights and chivalry).

MOCK TEST (WITH ANSWERS)

1. "The Age of Chivalry" is written by:

A. John Keats

B. George Krokos ✓

C. William Wordsworth

D. Robert Frost

Explanation: George Krokos. [easy, web-research]

2. The main theme of "The Age of Chivalry" is:

A. Friendship

B. Nature

C. Adventure and Chivalry ✓

D. Romance alone

Explanation: Adventure and Chivalry. [easy, web-research]

3. According to the poem, knights fought mainly for:

A. Money

B. Revenge

C. Show, love and glory ✓

D. Peace

Explanation: Show, love, and glory. [easy, web-research]

4. "Shining armour" in the poem represents:

A. Cowardice

B. Glory and honour ✓

C. Metal

D. Wealth

Explanation: Glory and honour. [medium, web-research]

5. Identify the figure of speech in "adventures hide themselves."

A. Simile

B. Metaphor

C. Personification ✓

D. Alliteration

Explanation: Personification — adventures are given the human ability to 'hide'. [medium, web-research]

6. Identify the figure of speech in "Maybe life itself is one big adventure."

A. Metaphor ✓

B. Simile

C. Irony

D. Personification

Explanation: Metaphor — life is directly equated with adventure. [medium, web-research]

7. In the poem, "transparent" is used to mean:

A. Invisible

B. Open to experience ✓

C. Weak

D. Lost

Explanation: Open to experience. [medium, web-research]

8. "Days of yore" refers to:

A. Modern age

B. Ancient times ✓

C. The future

D. Teenage years

Explanation: Ancient times. [easy, web-research]